Track 2 – The Shadow Wall: Attenuation

Objective

The objective of this project is to study how an electromagnetic or optical signal weakens when it passes through a shielding wall or an attenuating medium.

To describe this phenomenon, we use the Beer–Lambert law, which shows that the signal intensity decreases as it travels through the material. The problem here is to determine whether a digital code transmitted from behind this barrier can still be recovered after passing through the material. This project demonstrates how attenuation models can be used to evaluate signal loss and the feasibility of decoding it in practical shielding systems.

Mathematical Model

When a wave penetrates an absorbing medium, its intensity decreases exponentially with depth according to the Beer–Lambert law:

$$I(z) = I_0 e^{-\alpha z}$$

where I_0 is the initial intensity, $I(z)$ is the intensity after traveling a distance z inside the material, and a is the attenuation coefficient.

For conductive shielding materials, attenuation can also be interpreted using the skin depth δ,which represents the characteristic penetration distance of the signal inside the material:

$$\delta = \sqrt{\frac{2}{\omega\mu\sigma}}$$

where $\omega = 2\pi f$ is the angular frequency, μ is the magnetic permeability of the material, and σ is its electrical conductivity.

A practical engineering rule is:

If $d > 5\delta$, the signal becomes too weak to be physically recovered.

This means that once the wall thickness d exceeds five times the skin depth, the transmitted code is effectively lost.

Vectorization / Physical Interpretation

The attenuation process can be interpreted geometrically as an exponential decay of signal amplitude along the penetration axis z. Instead of tracking every electromagnetic interaction inside the material, the model summarizes the loss with a compact scalar expression. This makes it efficient for computer simulations and real-time physical validation.

➢ Test Case Validation

We tested the model using the following example values:

- Frequency: f = 1 MHz
- Material conductivity: σ = 5.8 × 107 S/m (copper-like shielding)
- Magnetic permeability: $\mu = \mu_0$= 4p × 10-7 H/m
- Wall thickness: d = 0.5 mm

First, we compute the angular frequency:

$$\omega = 2\pi f = 2\pi \times 10^6$$

Then the skin depth is approximately:

$$\delta = 6.6 \times 10\text{-}5 \text{ m} = 0.066 \text{ mm}$$

Now compare the wall thickness to 5d:

$$5\delta = 0.33 \text{ mm}$$

Since d = 0.5 mm > 0.33 mm, the condition d > 5δ is satisfied.

Result: The transmitted digital code cannot be physically recovered after crossing the shielding wall.

➢ Implementation

The algorithm follows these steps:

1. Input the signal frequency f, the material properties μ and σ, and the wall thickness d.
2. Compute the angular frequency: ω = 2πf.
3. Compute the skin depth: $\delta = \sqrt{\frac{2}{\omega\mu\sigma}}$
4. Compare the partition thickness with the threshold 5δ.
5. If d > 5δ, declare the signal unrecoverable; otherwise, decoding remains physically possible.
6. Optionally compute the residual intensity using $I(d) = I_0 e^{-\alpha d}$

The algorithm has constant complexity **O(1)**, because it only performs a fixed number of mathematical operations.

➢ Visualization

The simulation can display a source, a shielding wall, and the attenuated signal propagation across the barrier. The scene typically includes:

- A signal source placed before the wall.
- A shaded rectangular partition representing the absorbing material.
- A colored wave or ray entering the wall and decreasing in amplitude.
- A reduced output signal behind the wall.
- A threshold indicator showing whether the remaining signal is decodable or not.

Physical Behavior:

As the signal enters the wall, its energy is progressively absorbed. Thin or weakly conductive materials allow part of the wave to survive, while thick or highly conductive materials suppress it rapidly. At sufficiently large thicknesses, the received intensity falls below the decoding threshold, so the code can no longer be extracted.

When the simulation reaches the condition **d > 5**δ, the result can be interpreted as a “signal lock” that remains closed: the shield successfully hides the transmitted information.

➢ Discussion

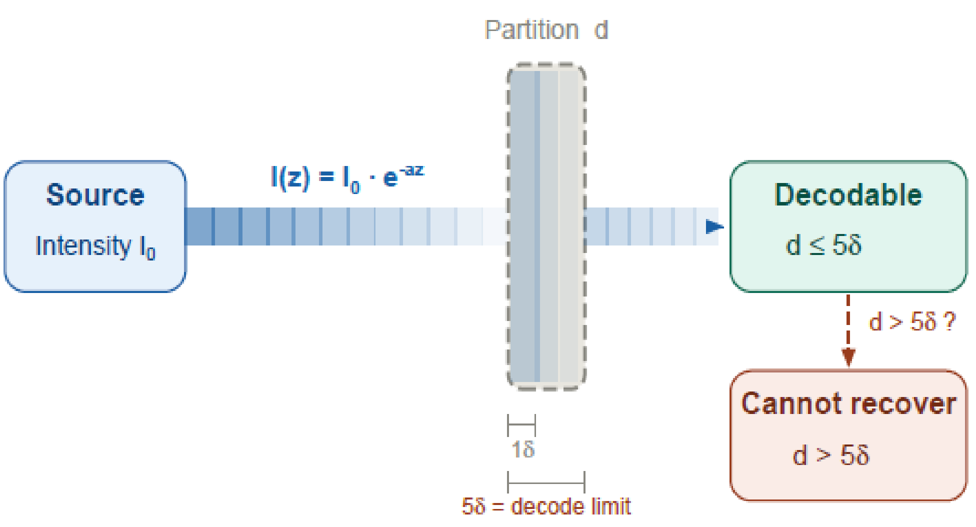
The simulation correctly models the attenuation of a signal through a shielding wall using exponential decay and skin-depth analysis. The method is physically meaningful because it connects measurable material properties (conductivity, permeability, and frequency) with a clear engineering decision: can the signal still be decoded or not?

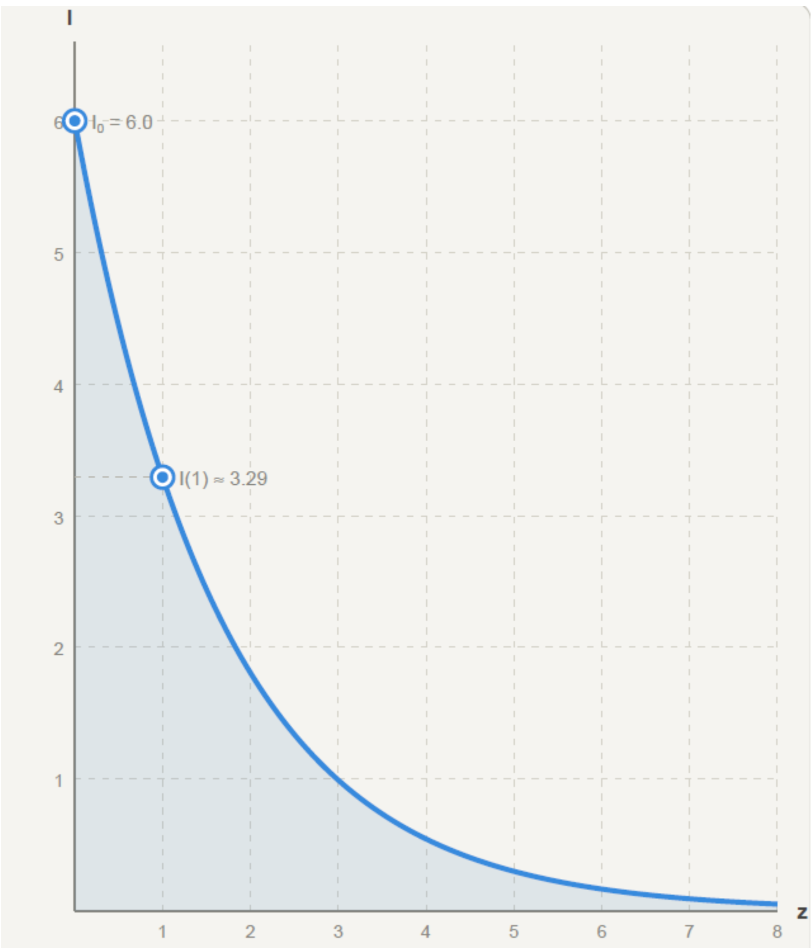
This principle is widely used in:

- Electromagnetic shielding and secure enclosures
- Radio-frequency interference protection
- Optical filtering and absorbing media
- Secure communications and anti-eavesdropping systems
- Sensor isolation and laboratory shielding

Small numerical variations may appear depending on unit conversions and floating-point precision, but the physical conclusion remains robust. The model is simple, efficient, and suitable for educational simulations in optics, waves, and computer-vision-inspired physics projects.

- Beer–Lambert / Attenuation Schema





This figure illustrates the signal attenuation phenomenon according to the Beer–Lambert law.

It can be observed that the signal intensity decreases exponentially as it passes through the medium.

The greater the thickness z, the more rapidly the intensity $I(z)$ decreases, which explains the difficulty of decoding when the signal becomes too weak.